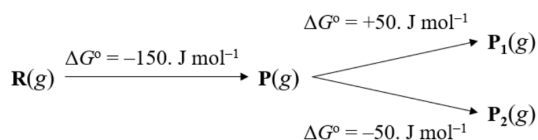


Thermodynamics

1. Substance $R(g)$ reacts to form $P(g)$, which then may isomerize to either $P_1(g)$ or $P_2(g)$. The diagram below shows ΔG° at 25°C .



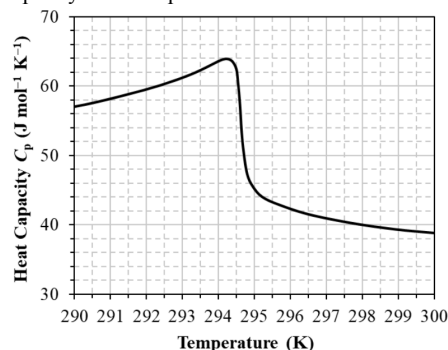
In an experiment, 10.0 g of $R(g)$ is placed in a sealed vessel. Which are true at equilibrium?

- I. The mass of $R(g)$ is 0.0 g.
 II. The mass of $P_1(g)$ is less than the mass of $P_2(g)$.
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
2. 60.0 mL of 0.60 M HCOOH and 40.0 mL of 0.80 M CH_3NH_2 , both initially at 22.0°C , are mixed in a well-insulated container. The final temperature is 27.9°C . What is $\Delta H^\circ_{\text{rxn}}$ for the reaction of formic acid and methylamine? Assume that the densities and specific heat capacities of all solutions are the same as pure water.
- $$\text{HCOOH}(aq) + \text{CH}_3\text{NH}_2(aq) \rightarrow \text{HCOO}^-(aq) + \text{CH}_3\text{NH}_3^+(aq) \quad \Delta H^\circ_{\text{rxn}} = ???$$
- (A) -31 kJ mol^{-1} (B) -41 kJ mol^{-1}
 (C) -69 kJ mol^{-1} (D) -77 kJ mol^{-1}
3. What is S° for $\text{O}_2(g)$?

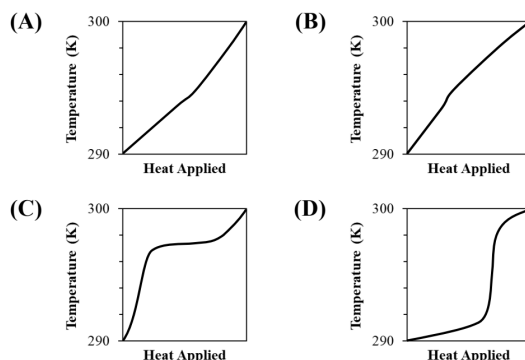
Species	ΔH°_f (kJ mol^{-1})	S° ($\text{J mol}^{-1} \text{K}^{-1}$)	ΔG°_f 298 K (kJ mol^{-1})
$\text{H}_2\text{O}(l)$	-290	70.	-240
$\text{H}_2\text{O}_2(l)$	-190	110.	-120

- (A) $150 \text{ J mol}^{-1} \text{K}^{-1}$ (B) $210 \text{ J mol}^{-1} \text{K}^{-1}$
 (C) $250 \text{ J mol}^{-1} \text{K}^{-1}$ (D) $290 \text{ J mol}^{-1} \text{K}^{-1}$

4. Between 293 K and 295 K, $\text{Gd}(s)$ transitions from a ferromagnetic to a paramagnetic state, causing a significant variation of the constant-pressure heat capacity with temperature as shown below.



A 10.0 g sample of $\text{Gd}(s)$, initially at 290 K, is gradually heated at a constant rate. Which curve best shows how temperature changes?



5. Which species has the greatest molar entropy S° at 298 K?
- (A) $\text{H}_2(g)$ (B) $\text{He}(g)$
 (C) $\text{Cl}_2(g)$ (D) $\text{Xe}(g)$
6. In an adiabatic process, no heat is exchanged with the surroundings. Which must be true in a reversible adiabatic process?
- I. $\Delta H = 0$ II. $\Delta S = 0$
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

Thermodynamics

Questions 7 and 8 use the following thermodynamic data:

Substance	ΔH°_f , kJ mol ⁻¹	ΔG°_f (298 K), kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
Cl ₂ (g)	0	0	223.1
PCl ₃ (g)	-287.0	-267.8	311.8
PCl ₅ (g)	-374.9		364.4

7. What is K_p for the gas-phase reaction of PCl₃ with Cl₂ at 450 K?



- (A) 19.8 (B) 2.02×10^4
(C) 3.17×10^6 (D) 1.60×10^{10}

8. What is ΔG°_f of PCl₅ at 298 K?

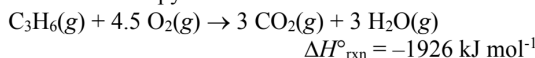
- (A) -266.3 kJ mol⁻¹ (B) -304.9 kJ mol⁻¹
(C) -355.7 kJ mol⁻¹ (D) -374.9 kJ mol⁻¹

9. Which quantities are independent of the path taken, given the same initial and final state?

- I. The work done by the reaction of a mole of gaseous propane with excess gaseous oxygen to form carbon dioxide and water vapor at a constant pressure of 1.0 atm
II. The heat absorbed by the expansion of a mole of carbon dioxide from a pressure of 4.0 atm to a pressure of 1.0 atm

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

10. The standard enthalpy of combustion of propene, C₃H₆(g), is -1926 kJ mol⁻¹. Given the tabulated bond dissociation enthalpies (BDEs), what is the bond dissociation enthalpy of a C=C bond?



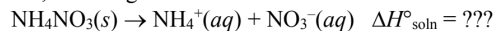
Bond	BDE, kJ mol ⁻¹	Bond	BDE, kJ mol ⁻¹
C-H	415	C-O	350
O-H	464	C=O	804
O-O	140	C-C	345
O=O	498	C=C	???

- (A) 476 kJ mol⁻¹ (B) 606 kJ mol⁻¹
(C) 690 kJ mol⁻¹ (D) 715 kJ mol⁻¹

11. Which reaction has $\Delta S^\circ > 0$?

- (A) Ga(l) → Ga(s)
(B) C₆₀(s) → 60 C(s, graphite)
(C) CaCO₃(s) → Ca²⁺(aq) + CO₃²⁻(aq)
(D) NH₄⁺(aq) + OH⁻(aq) → NH₃(aq) + H₂O(l)

12. A sample of 10.0 g ammonium nitrate ($M = 80.1$) is added to 100.0 g water in a well-insulated container. The solid and liquid are both initially at 22.0 °C, and the temperature of the solution after the ammonium nitrate has dissolved is 15.1 °C. What is the enthalpy of solution of ammonium nitrate? You may assume that the specific heat capacity of the solution is the same as that of pure water, 4.184 J g⁻¹ K⁻¹.



- (A) -3.2 kJ mol⁻¹ (B) 3.2 kJ mol⁻¹
(C) 5.5 kJ mol⁻¹ (D) 25 kJ mol⁻¹

13. Steam turbines convert heat energy into mechanical work. Which statement best describes the efficiency of this process?

- (A) 100%, because energy is conserved
(B) Less than 100%, because the steam decreases in temperature
(C) Greater than 100%, because the turbine increases in kinetic energy
(D) Can be greater than, less than, or equal to 100% depending on the design of the turbine.

14. At what temperatures is the decomposition of tungstic acid, H₂WO₄(s), to tungsten trioxide and water vapor spontaneous under standard conditions?



Compound	ΔH°_f , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
H ₂ WO ₄ (s)	-1130	140
WO ₃ (s)	-840	80
H ₂ O(g)	-240	190

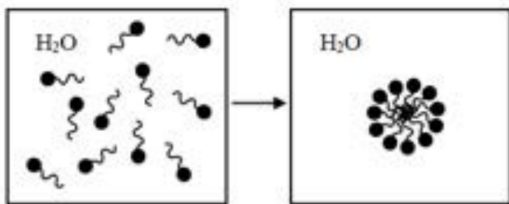
- (A) $T > 385 \text{ K}$
(B) $T < 385 \text{ K}$
(C) The reaction is spontaneous at all temperatures.
(D) The reaction is not spontaneous at any temperature.

15. In a well-insulated container, 10.0 g solid octane at its freezing point (-56.9 °C) is added to 300.0 mL liquid octane at 0.0 °C (density = 0.71 g mL⁻¹). After equilibrium is achieved, the temperature is -6.3 °C. When the experiment is repeated under the same conditions except with 100.0 g solid octane, what is the final temperature?

- (A) -39.7 °C (B) -44.9 °C
(C) -51.7 °C (D) -56.9 °C

Thermodynamics

16. Amphiphilic species like $\text{CH}_3(\text{CH}_2)_{10}\text{COO}^-$ can aggregate into micelles in water, as depicted schematically below. ΔS for micelle formation is positive. Which best explains why?



- (A) Amphiphilic compounds are solvated by an ordered layer of water molecules.
 (B) Micelle formation involves converting many independent amphiphilic compounds into a single large micelle.
 (C) Water molecules are trapped in the hydrophobic interior of micelles.
 (D) Water molecules form ordered structures around the exterior of micelles.

17. What is ΔG°_f of $\text{C}_2\text{H}_4(\text{g})$ at 25 °C?

Substance	ΔH°_f , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
$\text{C}_2\text{H}_4(\text{g})$	52.4	219.3
$\text{H}_2(\text{g})$	0	130.7
$\text{C}(\text{s, graphite})$	0	5.7

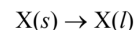
- (A) $-13.0 \text{ kJ mol}^{-1}$ (B) 27.7 kJ mol^{-1}
 (C) 46.9 kJ mol^{-1} (D) 68.3 kJ mol^{-1}
18. The ΔH°_f of $\text{CH}_4(\text{g})$ is $-74.6 \text{ kJ mol}^{-1}$ and the bond dissociation enthalpies (BDEs) of several bonds are given in the table.

Bond	BDE, kJ mol ⁻¹
C–C	347
C–H	413
H–H	432

Based on these data, what is the molar enthalpy of sublimation of $\text{C}(\text{s, graphite})$ to form $\text{C}(\text{g})$?

- (A) 713 kJ mol^{-1} (B) 788 kJ mol^{-1}
 (C) 1061 kJ mol^{-1} (D) 1135 kJ mol^{-1}

19. The phase change



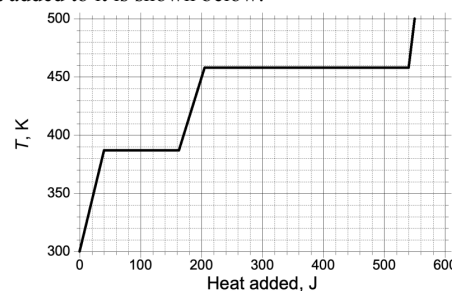
has $\Delta G^\circ > 0$ at 300 K. Which statement about this reaction must be correct?

- (A) $K_{\text{eq}} < 1$ at 300 K
 (B) $\Delta S^\circ < 0$ at 300 K
 (C) Liquid X cannot exist at 300 K
 (D) The melting point of X is greater than 300 K

20. For which reaction is $\Delta H^\circ_{\text{rxn}}$ closest to $\Delta G^\circ_{\text{rxn}}$?

- (A) $2 \text{ CO}_2(\text{g}) \rightarrow 2 \text{ CO}(\text{g}) + \text{O}_2(\text{g})$
 (B) $2 \text{ HCl}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{Cl}_2(\text{g})$
 (C) $2 \text{ H}_2\text{O}(\text{l}) \rightarrow 2 \text{ H}_2(\text{g}) + \text{O}_2(\text{g})$
 (D) $2 \text{ NaCl}(\text{s}) \rightarrow 2 \text{ Na}(\text{s}) + \text{Cl}_2(\text{g})$

21. 2.00 g solid iodine, initially at 300 K, is heated in a sealed vessel maintained at 1.00 atm pressure. The temperature of the sample as a function of the amount of heat added to it is shown below.

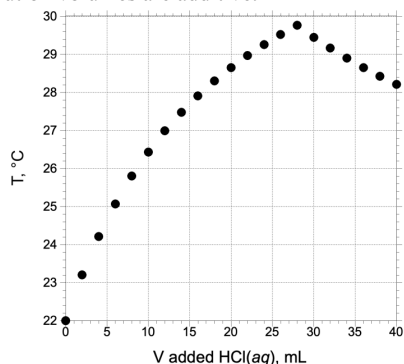


What is the entropy of vaporization of I_2 , $\Delta S^\circ_{\text{vap}}$, at its normal boiling point?

- (A) $40.1 \text{ J mol}^{-1} \text{ K}^{-1}$ (B) $52.6 \text{ J mol}^{-1} \text{ K}^{-1}$
 (C) $56.8 \text{ J mol}^{-1} \text{ K}^{-1}$ (D) $92.8 \text{ J mol}^{-1} \text{ K}^{-1}$

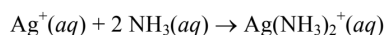
Thermodynamics

22. 20.0 mL of NaOH(aq) is titrated against 1.00 M HCl(aq) in a well-insulated vessel with constant stirring. Each solution is initially at 22.0° C, and the temperature of the reaction mixture is monitored after each 2.0 mL of hydrochloric acid is added. What is ΔH° for the reaction that takes place? Assume that all solutions' densities and heat capacities are the same as that of pure water, and that solution volumes are additive.



- (A) -15 kJ mol^{-1} (B) -31 kJ mol^{-1}
(C) -39 kJ mol^{-1} (D) -54 kJ mol^{-1}

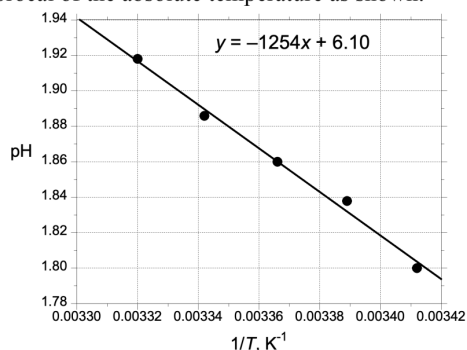
23. Silver ion reacts with ammonia to form a complex ion:



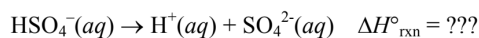
In a certain solution at equilibrium at 298 K, $[\text{Ag}^+] = [\text{NH}_3] = [\text{Ag}(\text{NH}_3)_2^+] = C$. What is ΔG_{rxn} at 298 K if $[\text{Ag}^+] = [\text{NH}_3] = [\text{Ag}(\text{NH}_3)_2^+] = 0.1 \text{ C}$?

- (A) -5.7 kJ mol^{-1} (B) 0 kJ mol^{-1}
(C) 5.7 kJ mol^{-1} (D) 11.4 kJ mol^{-1}

24. Equal volumes of 1.00 M $\text{HSO}_4^-(aq)$ and 1.00 M $\text{SO}_4^{2-}(aq)$ are mixed. The pH of this solution varies with the reciprocal of the absolute temperature as shown.



What is the standard enthalpy change for the acid dissociation of $\text{HSO}_4^-(aq)$?



- (A) $-24.0 \text{ kJ mol}^{-1}$ (B) $-10.4 \text{ kJ mol}^{-1}$
(C) 10.4 kJ mol^{-1} (D) 24.0 kJ mol^{-1}

25. When a helium-filled balloon is immersed in liquid nitrogen, which statements accurately describe the changes as the balloon deflates?

- I. The nitrogen does work on the balloon.
II. The entropy of the nitrogen increases.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

26. Solid KOH ($M = 56.11$, $\Delta H^\circ_{\text{soln}} = -57.6 \text{ kJ mol}^{-1}$) is slowly added to distilled water contained in a well-insulated vessel, initially at 25.0 °C. What is the pH when the temperature of the solution reaches 26.0 °C? You may assume that the density of the solution remains at 1.00 g mL^{-1} and its heat capacity remains at $4.184 \text{ J g}^{-1} \text{ K}^{-1}$ throughout the experiment.

- (A) 9.1 (B) 9.9 (C) 12.1 (D) 12.9

27. What is ΔG°_f (at 298 K) for $\text{ClF}_3(g)$?

Substance	$\Delta H^\circ_f, \text{ kJ mol}^{-1}$	$S^\circ, \text{ J mol}^{-1} \text{ K}^{-1}$
$\text{F}_2(g)$	0	202.8
$\text{Cl}_2(g)$	0	223.1
$\text{ClF}_3(g)$	-158.9	281.6

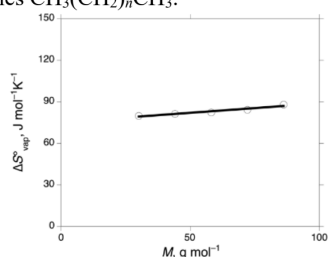
- (A) $-78.9 \text{ kJ mol}^{-1}$ (B) $-83.9 \text{ kJ mol}^{-1}$
(C) $-118.9 \text{ kJ mol}^{-1}$ (D) $-242.8 \text{ kJ mol}^{-1}$

28. The triple point of bromine is 265.9 K and 0.058 bar, and the standard entropy of vaporization of bromine is $89.0 \text{ J mol}^{-1} \text{ K}^{-1}$. What is the standard enthalpy of vaporization of bromine?

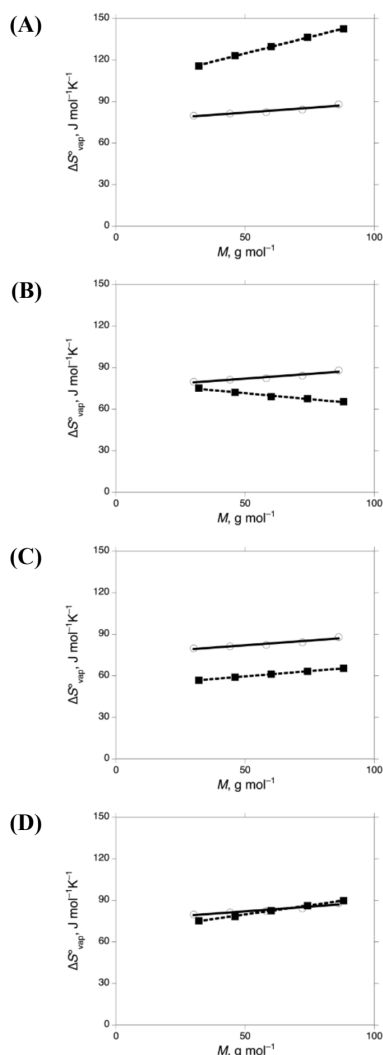
- (A) 17.4 kJ mol^{-1} (B) 23.7 kJ mol^{-1}
(C) 26.5 kJ mol^{-1} (D) 30.0 kJ mol^{-1}

Thermodynamics

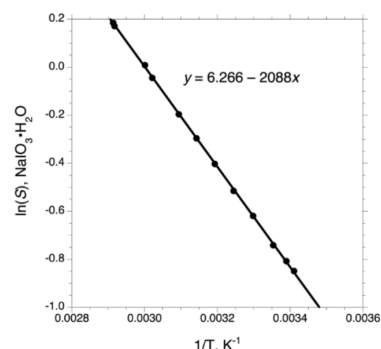
29. The graph below shows how the standard entropy of vaporization, $\Delta S^\circ_{\text{vap}}$, varies with molar mass for straight-chain alkanes $\text{CH}_3(\text{CH}_2)_n\text{CH}_3$.



Which of the following graphs best shows how $\Delta S^\circ_{\text{vap}}$ (solid squares, dashed line) varies with molar mass for straight-chain primary alcohols $\text{CH}_3(\text{CH}_2)_n\text{OH}$?



30. The natural logarithm of the molar solubility S of sodium iodate monohydrate as a function of the reciprocal of the absolute temperature is plotted below.



What is the standard enthalpy of dissolution for $\text{NaIO}_3 \cdot \text{H}_2\text{O}$?

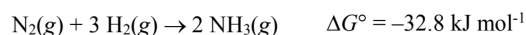
- (A) 2.1 kJ mol^{-1} (B) 17.4 kJ mol^{-1}
(C) 26.0 kJ mol^{-1} (D) 34.7 kJ mol^{-1}
31. An insulated container holds 100.0 g of water at 25.0 °C. A 100.0 g piece of copper at 50.0 °C is added to the container. What is the final temperature of the mixture in the container at equilibrium? The specific heat capacity of copper is 0.385 $\text{J g}^{-1}\text{K}^{-1}$.
- (A) 27.1 °C (B) 27.5 °C
(C) 34.9 °C (D) 37.5 °C
32. At the triple point of water (0.01 °C, 0.006 atm), in which phase would 1.0 mol of water have the lowest entropy?
- (A) Solid
(B) Liquid
(C) Gas
(D) It would have the same entropy in all three phases.
33. The standard enthalpy of formation of $\text{NO}_2(\text{g})$ is 33.1 kJ mol^{-1} . What is the average bond dissociation enthalpy of a nitrogen-oxygen bond in $\text{NO}_2(\text{g})$?

Bond	Bond dissociation enthalpy, kJ mol^{-1}
$\text{N}\equiv\text{N}$	945.4
$\text{O}=\text{O}$	498.4

- (A) 469.0 kJ mol^{-1} (B) 502.1 kJ mol^{-1}
(C) 705.4 kJ mol^{-1} (D) 738.5 kJ mol^{-1}

Thermodynamics

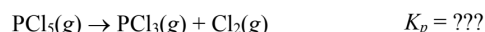
34. The formation of ammonia from nitrogen and hydrogen has a standard free energy change of $-32.8 \text{ kJ mol}^{-1}$ at 298 K.



What is ΔG at 298 K if the partial pressures of all three gases are 3.00 bar?

- (A) $-30.1 \text{ kJ mol}^{-1}$ (B) $-32.8 \text{ kJ mol}^{-1}$
(C) $-35.5 \text{ kJ mol}^{-1}$ (D) $-38.2 \text{ kJ mol}^{-1}$

35. What is K_p for the dissociation of $\text{PCl}_5(\text{g})$ at 298 K?



Substance	$\Delta G^\circ_f, \text{ kJ mol}^{-1}$
$\text{PCl}_5(\text{g})$	-325
$\text{PCl}_3(\text{g})$	-286

- (A) 1.8×10^{-16} (B) 5.9×10^{-11}
(C) 6.0×10^{-9} (D) 1.5×10^{-7}

36. A reaction has $K_{\text{eq}} = 0.10$ at 300 K and $K_{\text{eq}} = 0.50$ at 330 K. Which statements about this reaction in this temperature range must be correct?

- I. $\Delta H^\circ_{\text{rxn}} > 0$
II. $\Delta S^\circ_{\text{rxn}} > 0$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

37. What is the standard enthalpy of formation of $\text{C}_2\text{H}_2(\text{g})$?

	$\Delta H^\circ_{\text{rxn}}, \text{ kJ mol}^{-1}$
$\text{C}_2\text{H}_2(\text{g}) + \frac{5}{2} \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	-1229.6
$\text{C}(\text{s, graphite}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$	-393.5
$\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$	-285.9

- (A) $-2302.5 \text{ kJ mol}^{-1}$ (B) $-1909.0 \text{ kJ mol}^{-1}$
(C) $156.7 \text{ kJ mol}^{-1}$ (D) $550.2 \text{ kJ mol}^{-1}$

38. Which reaction occurs with the greatest increase in entropy?

- (A) $\text{ClF}(\text{s}) \rightarrow \text{ClF}(\text{g})$
(B) $\text{ClF}(\text{s}) \rightarrow \text{ClF}(\text{l})$
(C) $\text{ClF}(\text{l}) \rightarrow \text{ClF}(\text{g})$
(D) $\text{ClF}(\text{g}) \rightarrow \frac{1}{2} \text{Cl}_2(\text{g}) + \frac{1}{2} \text{F}_2(\text{g})$

39. A 2.00 g sample of solid RbClO_4 ($M = 184.92$) is added to 100.0 g water, both initially at 23.00°C , in a well-insulated container. The final temperature of the solution is 21.56°C . What is $\Delta H^\circ_{\text{solution}}$ of RbClO_4 ? (Assume that the specific heat capacity of the solution is the same as that of pure water and neglect the heat capacity of the insulated container.)

- (A) $-0.615 \text{ kJ mol}^{-1}$ (B) $0.615 \text{ kJ mol}^{-1}$
(C) 13.6 kJ mol^{-1} (D) 56.8 kJ mol^{-1}

40. For a reaction studied over a wide temperature range, plots of $\Delta G^\circ_{\text{rxn}}/T$ vs. $1/T$ and of $\Delta G^\circ_{\text{rxn}}$ vs. T are linear. Which of the following must be independent of temperature?

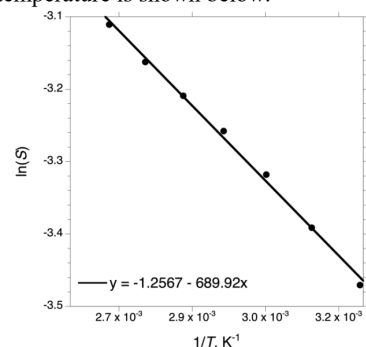
- I. $\Delta G^\circ_{\text{rxn}}$
II. $\Delta H^\circ_{\text{rxn}}$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

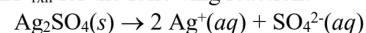
41. Elemental tin forms two allotropes, white tin and gray tin. For the conversion of white tin to gray tin, $\Delta H^\circ = -2.02 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = -7.04 \text{ J mol}^{-1} \text{ K}^{-1}$. Which statement best describes the stability of the two phases?

- (A) White tin is stable below 13.8°C and gray tin is stable above this temperature.
(B) Gray tin is stable below 13.8°C and white tin is stable above this temperature.
(C) White tin is stable below 0.3°C and gray tin is stable above this temperature.
(D) Gray tin is stable below 0.3°C and white tin is stable above this temperature.

42. The plot of the natural logarithm of the molar solubility (S) of Ag_2SO_4 as a function of the reciprocal of the absolute temperature is shown below.



What is $\Delta H^\circ_{\text{rxn}}$ for the following reaction?



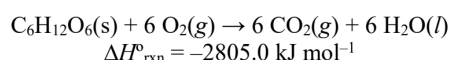
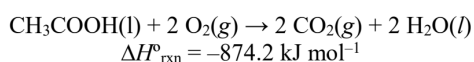
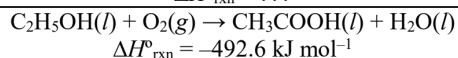
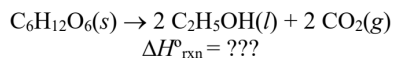
- (A) $-19.8 \text{ kJ mol}^{-1}$ (B) $-10.5 \text{ kJ mol}^{-1}$
(C) 5.74 kJ mol^{-1} (D) 17.2 kJ mol^{-1}

Thermodynamics

43. The process of liquid water freezing at -10°C and 1 atm pressure is

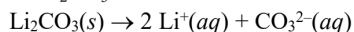
(A) spontaneous and exothermic
(B) spontaneous and endothermic
(C) nonspontaneous and exothermic
(D) nonspontaneous and endothermic

44. What is $\Delta H^\circ_{\text{rxn}}$ for the fermentation of glucose as shown below?



(A) $-71.4 \text{ kJ mol}^{-1}$ (B) $-945.6 \text{ kJ mol}^{-1}$
(C) $-1438.2 \text{ kJ mol}^{-1}$ (D) $-5528.7 \text{ kJ mol}^{-1}$

45. The solubility of Li_2CO_3 in water at 298 K is 0.175 mol L^{-1} , and its solubility decreases with increasing temperature. What are the signs of ΔH° and ΔS° for the dissolution of Li_2CO_3 ?

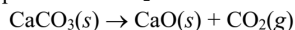


(A) $\Delta H^\circ > 0, \Delta S^\circ > 0$ (B) $\Delta H^\circ > 0, \Delta S^\circ < 0$
(C) $\Delta H^\circ < 0, \Delta S^\circ > 0$ (D) $\Delta H^\circ < 0, \Delta S^\circ < 0$

46. A system confined in a rigid, well-insulated container undergoes a spontaneous change. What statement about the system must be true?

(A) Its Gibbs free energy increases during the change.
(B) Its Gibbs free energy decreases during the change.
(C) Its entropy increases during the change.
(D) Its entropy decreases during the change.

47. What is ΔG for the decomposition of CaCO_3 at 298 K and a partial pressure of CO_2 of $4.00 \times 10^{-4} \text{ bar}$?



$$\Delta G_{\text{rxn}}(298 \text{ K}, P_{\text{CO}_2} = 0.400 \text{ mbar}) = ???$$

Compound	$\text{CaCO}_3(s)$	$\text{CaO}(s)$	$\text{CO}_2(g)$
$\Delta G^\circ_f(\text{kJ mol}^{-1})$	-1129	-604	-394

(A) -131 kJ mol^{-1} (B) 112 kJ mol^{-1}
(C) 131 kJ mol^{-1} (D) 150 kJ mol^{-1}

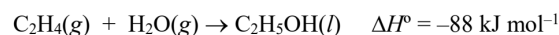
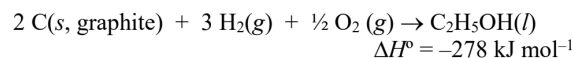
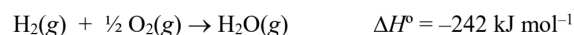
48. A 10.0 g sample of solid NH_4NO_3 ($M = 80.05$) is added to 100.0 g H_2O in a well-insulated container. Both the solid and the water are initially at 24.00°C , but after the NH_4NO_3 dissolves, the temperature of the mixture is 17.11°C . What is $\Delta H^\circ_{\text{rxn}}$ for the dissolution of $\text{NH}_4\text{NO}_3(s)$? Assume that the specific heat capacity of the NH_4NO_3 solution is the same as that of pure water, $4.184 \text{ J g}^{-1}^\circ\text{C}^{-1}$.

(A) $-3.17 \text{ kJ mol}^{-1}$ (B) $0.559 \text{ kJ mol}^{-1}$
(C) 3.17 kJ mol^{-1} (D) 25.4 kJ mol^{-1}

49. A spontaneous endothermic chemical reaction takes place at constant pressure and volume in a well-insulated container. As the reaction proceeds, what happens to the temperature of the system and to the internal energy of the reactants as they are transformed into products?

(A) Both the temperature and the reactants' internal energy increase.
(B) The temperature increases while the reactants' internal energy decreases.
(C) The temperature decreases while the reactants' internal energy increases.
(D) Both the temperature and the reactants' internal energy decrease.

50. What is the standard heat of formation (ΔH°_f) of gaseous ethylene, $\text{C}_2\text{H}_4(g)$?



(A) 16 kJ mol^{-1} (B) 52 kJ mol^{-1}
(C) 83 kJ mol^{-1} (D) 285 kJ mol^{-1}

51. The K_{sp} of AgBr is 5.0×10^{-13} at 25.0°C and 6.5×10^{-12} at 50.0°C . What is $\Delta H^\circ_{\text{rxn}}$ for the precipitation of AgBr shown?



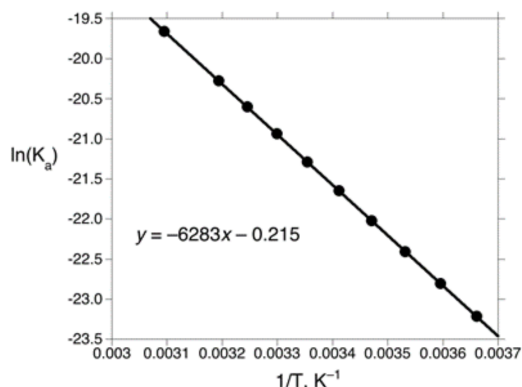
(A) -82 kJ mol^{-1} (B) -1.1 kJ mol^{-1}
(C) 1.1 kJ mol^{-1} (D) 82 kJ mol^{-1}

Thermodynamics

52. For benzene, C_6H_6 , the heat of fusion is 9.9 kJ mol^{-1} at its melting point of 5°C , its heat of vaporization is 30.8 kJ mol^{-1} at its normal boiling point of 80°C , and the molar heat capacity of the liquid is $135 \text{ J mol}^{-1} \text{ K}^{-1}$. To a system consisting of 1.00 mol pure solid C_6H_6 at 5°C is added 20.0 kJ of heat while maintaining the pressure at 1 atm . In what phases is the benzene found?

(A) Mixture of solid and liquid
(B) Liquid only
(C) Mixture of liquid and gas
(D) Gas only

53. The natural logarithm of the K_a of NH_4^+ is plotted below as a function of the reciprocal of the absolute temperature.



Which statements are correct?

- I. The pH of a 0.100 M solution of NH_4NO_3 increases as the temperature is raised.
II. ΔS° for the acid dissociation of ammonium ion is positive.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

54. The molar heat capacity of ice is $37 \text{ J mol}^{-1} \text{ K}^{-1}$ and the molar heat capacity of liquid water is $76 \text{ J mol}^{-1} \text{ K}^{-1}$. What can be concluded about the relative magnitudes of the heat of fusion of ice at 263 K (ΔH_{263}) compared to the heat of fusion of ice at 273 K (ΔH_{273})?

(A) $\Delta H_{263} < \Delta H_{273}$
(B) $\Delta H_{263} = \Delta H_{273}$
(C) $\Delta H_{263} > \Delta H_{273}$
(D) No conclusion about the relative magnitudes of the heats of fusion can be drawn based on the information given.

55. What is ΔH° for the reaction shown?



Compound	$\Delta H_f^\circ, \text{ kJ mol}^{-1}$
$\text{H}_2\text{S}(\text{g})$	-20.15
$\text{H}_2\text{O}(\text{l})$	-285.8
$\text{SO}_2(\text{g})$	-296.4

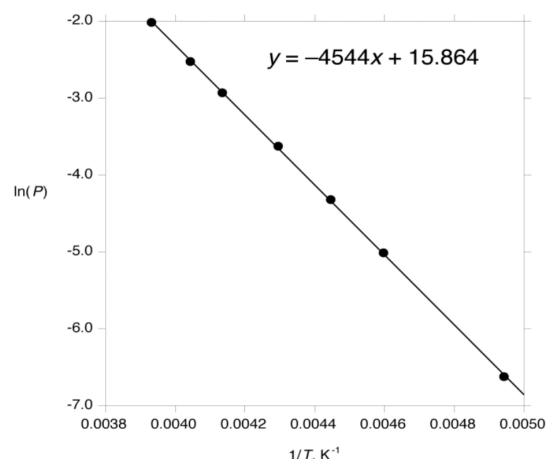
(A) $-19.4 \text{ kJ mol}^{-1}$ (B) $-374.7 \text{ kJ mol}^{-1}$
(C) $-562.1 \text{ kJ mol}^{-1}$ (D) $-1124.1 \text{ kJ mol}^{-1}$

56. Upon which factors can the Gibbs free energy change for a reaction (ΔG_{rxn}) depend?

- I. Temperature
II. Concentration of species in solution

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

57. The natural logarithm of the vapor pressure (in bar) of $\text{HCN}(\text{s})$ as a function of the reciprocal of the absolute temperature is plotted below.



What is the heat of sublimation of HCN ?

(A) 4.54 kJ mol^{-1} (B) 15.9 kJ mol^{-1}
(C) 37.8 kJ mol^{-1} (D) 132 kJ mol^{-1}

58. Which has the greatest entropy at 0°C ?

(A) $1.0 \text{ mol H}_2\text{O}(\text{s})$
(B) $1.0 \text{ mol H}_2\text{O}(\text{l})$
(C) $1.0 \text{ mol H}_2\text{O}(\text{g})$
(D) $1.0 \text{ mol H}_2(\text{g}) + 0.5 \text{ mol O}_2(\text{g})$

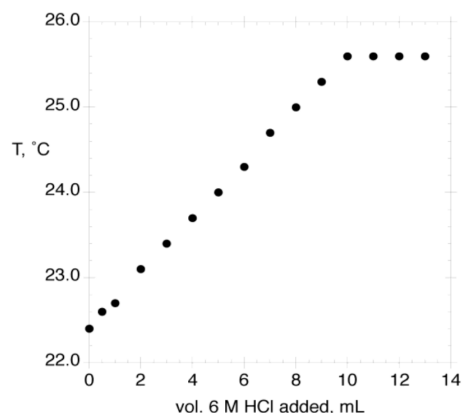
Thermodynamics

59. What is ΔG°_f of $\text{CH}_4(g)$ at 298 K? (All data are given at 298 K.)

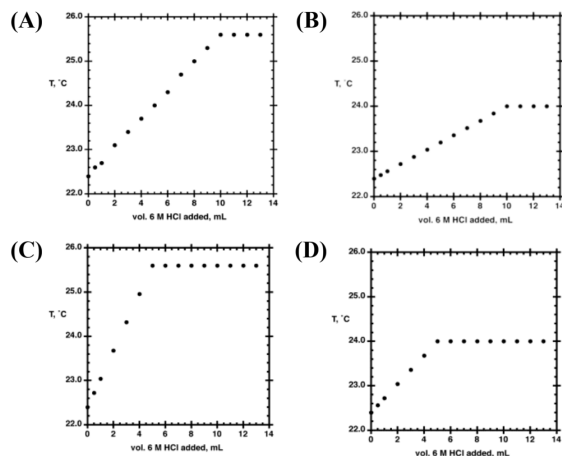
Substance	ΔH°_f , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹	ΔG°_f , kJ mol ⁻¹
$\text{CH}_4(g)$	-74.8	186.3	
$\text{H}_2(g)$	0.0	130.7	0.0
$\text{CO}_2(g)$	-393.5	213.7	-394.4
$\text{H}_2\text{O}(g)$	-241.8	188.8	-228.6

- (A) -50.7 kJ mol⁻¹ (B) -75.7 kJ mol⁻¹
(C) -98.0 kJ mol⁻¹ (D) -130.3 kJ mol⁻¹

60. A solution containing 0.060 mol NaOH dissolved in 200 g water was treated with successive aliquots of 6.0 M aqueous HCl in a well-insulated flask, and the temperature was measured after each aliquot to give the following data:



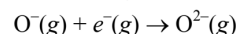
The same experiment was repeated, again using 0.060 mol NaOH and 6.0 M aqueous HCl, but this time the NaOH was initially dissolved in 400 g water. Which graph represents the data obtained in this experiment?



61. Given



What is ΔH° for the following reaction?



- (A) -844 kJ mol⁻¹ (B) -560 kJ mol⁻¹
(C) +560 kJ mol⁻¹ (D) +844 kJ mol⁻¹

62. For which reaction is $\Delta S^\circ < 0$?

- (A) $\text{NH}_4\text{Br}(s) \rightarrow \text{NH}_3(g) + \text{HBr}(g)$
(B) $\text{C}_2\text{H}_5\text{OH}(l) \rightarrow \text{C}_2\text{H}_5\text{OH}(g)$
(C) $2 \text{NO}_2(g) \rightarrow 2 \text{NO}(g) + \text{O}_2(g)$
(D) $2 \text{BrCl}(g) \rightarrow \text{Cl}_2(g) + \text{Br}_2(l)$

63. A hot metal coin is dropped into cold water in a well-insulated container. Which statements are true as the system approaches equilibrium?

I. $|q_{\text{metal}}| = |q_{\text{water}}|$

II. $|\Delta S_{\text{metal}}| = |\Delta S_{\text{water}}|$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

64. What is ΔG°_f of $\text{C}_6\text{H}_6(g)$ at 298 K?

	$\text{C}_6\text{H}_6(l)$	$\text{C}_6\text{H}_6(g)$
ΔH°_f , kJ mol ⁻¹	49.0	82.9
S° , J mol ⁻¹ K ⁻¹	173.3	269.0
ΔG°_f (298 K), kJ mol ⁻¹	124.3	

- (A) 2.7 kJ mol⁻¹ (B) 121.6 kJ mol⁻¹
(C) 127.0 kJ mol⁻¹ (D) 129.7 kJ mol⁻¹

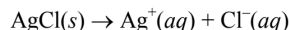
65. Assuming that ΔH° and ΔS° do not vary significantly with temperature, at what temperature will graphite and diamond be in equilibrium at 1 atm pressure?

	ΔH°_f , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
C (s, graphite)	0.0	5.9
C (s, diamond)	2.1	2.5

- (A) 0.62 K
(B) 620 K
(C) 2400 K
(D) Graphite is more stable than diamond at all temperatures.

Thermodynamics

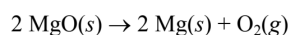
66. The K_{sp} of AgCl is 9.5×10^{-11} at 14°C and is 7.8×10^{-10} at 42°C . What is ΔH° for the dissolution of AgCl(s)?



- (A) $-0.49 \text{ kJ mol}^{-1}$ (B) $+0.37 \text{ kJ mol}^{-1}$
(C) $+57 \text{ kJ mol}^{-1}$ (D) $+220 \text{ kJ mol}^{-1}$
67. A system consists of a gas contained in a thin balloon. If the balloon deflates as the temperature of the gas changes from 90°C to 25°C , then

- (A) Heat is transferred out of the system and work is done on the system.
(B) Heat is transferred out of the system and work is done by the system.
(C) Heat is transferred into the system and work is done on the system.
(D) Heat is transferred into the system and work is done by the system.

68. The ΔH_f° of MgO is $-602 \text{ kJ}\cdot\text{mol}^{-1}$. When 20.15 g MgO is decomposed at constant pressure according to the equation below, how much heat will be transferred?

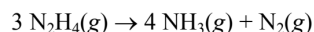


- (A) $1.20 \times 10^3 \text{ kJ}$ of heat is released
(B) $6.02 \times 10^2 \text{ kJ}$ of heat is absorbed
(C) $6.02 \times 10^2 \text{ kJ}$ of heat is released
(D) $3.01 \times 10^2 \text{ kJ}$ of heat is absorbed

69. What mass of ice at 0.0°C must be added to 100. g H_2O at 25.0°C to cool it to 0.0°C ? The heat of fusion of ice is $334 \text{ J}\cdot\text{g}^{-1}$.

- (A) 1.25 g (B) 7.49 g (C) 31.3 g (D) 100. g

70. Using the bond dissociation enthalpies (BDE) in the table, estimate ΔH° for the disproportionation of hydrazine described in the equation below.



Bond	BDE, $\text{kJ}\cdot\text{mol}^{-1}$	Bond	BDE, $\text{kJ}\cdot\text{mol}^{-1}$
N–N	163	N $\equiv$ N	944
N=N	409	N–H	388

- (A) $+283 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-283 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $-393 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $-455 \text{ kJ}\cdot\text{mol}^{-1}$

71. Which halogen has the highest standard entropy, S° ?

- (A) $\text{F}_2(g)$ (B) $\text{Cl}_2(g)$ (C) $\text{Br}_2(l)$ (D) $\text{I}_2(s)$

72. A chemical reaction has $K_{eq} = 1 \times 10^{-5}$ at 25°C , and the value of K_{eq} increases with increasing temperature. From these statements, what may one conclude?

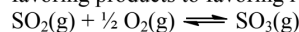
- (A) $\Delta H^\circ > 0$ and $\Delta S^\circ > 0$
(B) $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$
(C) $\Delta H^\circ < 0$ and $\Delta S^\circ > 0$
(D) $\Delta H^\circ > 0$ and no conclusion may be drawn about the sign of ΔS°

73. The combustion of 2-propanol ($M = 60.0 \text{ g}\cdot\text{mol}^{-1}$) occurs according to the equation,
 $2 \text{ CH}_3\text{CHOHCH}_3(l) + 9 \text{ O}_2(g) \rightarrow 6 \text{ CO}_2(g) + 8 \text{ H}_2\text{O}(l)$
What is q for the combustion of 15.0 g of 2-propanol?

	$\text{CH}_3\text{CHOHCH}_3(l)$	$\text{CO}_2(g)$	$\text{H}_2\text{O}(l)$
$\Delta H_f^\circ, \text{kJ}\cdot\text{mol}^{-1}$	-318.2	-393.5	-285.8

- (A) $-5.01 \times 10^2 \text{ kJ}$ (B) $-1.00 \times 10^3 \text{ kJ}$
(C) $-2.01 \times 10^3 \text{ kJ}$ (D) $-4.01 \times 10^3 \text{ kJ}$

74. At what temperature does the reaction below change from favoring products to favoring reactants?



	$\text{SO}_2(g)$	$\text{O}_2(g)$	$\text{SO}_3(g)$
$\Delta H_f^\circ, \text{kJ}\cdot\text{mol}^{-1}$	-296.8		-395.7
$S^\circ, \text{kJ}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$	0.248	0.205	0.257

- (A) 162 K (B) 509 K (C) 1060 K (D) 1540 K

75. Which isomer of C_4H_8 has the lowest absolute entropy at 25°C ?

- (A) 1-butene (B) *cis*-2-butene
(C) *trans*-2-butene (D) cyclobutane

76. A 10.0 g piece of gallium ($M = 69.7$) at 25.0°C is placed in 10.0 g of H_2O at 55.0°C . What is the final temperature when this system comes to equilibrium? (Assume the specific heat capacity of liquid Ga is the same as that of solid Ga.)

	$C_p, \text{J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$	melting point $^\circ\text{C}$	$\Delta H_{fus}, \text{kJ}\cdot\text{mol}^{-1}$
Ga	0.371	30.0	5.576
H_2O	4.184		

- (A) 35.0°C (B) 38.1°C
(C) 41.8°C (D) 52.6°C

77. In the Born-Haber calculation of the lattice enthalpy of LiF from its elements, which process is exothermic?

- (A) dissociation energy of $\text{F}_2(g)$
(B) electron affinity of F(g)
(C) ionization energy of Li(g)
(D) sublimation energy of Li(s)

Thermodynamics

78. How much work is done by the gas when 1.00 g of sodium azide, NaN_3 ($M = 65.01 \text{ g}\cdot\text{mol}^{-1}$), decomposes in a container of changeable volume (e.g. an airbag in a car) against a constant pressure of 1.00 atm at 298 K?
- $$2 \text{NaN}_3(\text{s}) \rightarrow 2 \text{Na}(\text{s}) + 3 \text{N}_2(\text{g})$$

(A) +114 J (B) +57.2 J (C) -114 J (D) -57.2 J

79. Which is (are) state properties?

I. enthalpy II. heat III. volume

(A) I. only (B) II. only
(C) I. and III. only (D) II. and III. only

80. $\text{HNO}_2(\text{l}) + \text{NaCl}(\text{s}) \rightarrow \text{HCl}(\text{g}) + \text{NaNO}_2(\text{s})$

Calculate the ΔH° value for the reaction above from the information below.

Reaction	$\Delta H^\circ \text{ kJ}\cdot\text{mol}^{-1}$
$\text{NO}(\text{g}) + \text{NO}_2(\text{g}) + \text{Na}_2\text{O}(\text{s}) \rightarrow 2\text{NaNO}_2(\text{s})$	-427.0
$\text{NO}(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{N}_2\text{O}(\text{g}) + \text{O}_2(\text{g})$	-43.0
$2\text{NaCl}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{HCl}(\text{g}) + \text{Na}_2\text{O}(\text{s})$	507.0
$2\text{HNO}_2(\text{l}) \rightarrow \text{N}_2\text{O}(\text{g}) + \text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	34.0

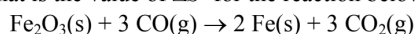
(A) 157 kJ (B) 78.5 kJ
(C) -78.5 kJ (D) -157 kJ

81. For which reaction(s) is the ΔH value close to the ΔE value?

I. $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \rightarrow \text{CO}(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$
II. $\text{C}_2\text{H}_4(\text{g}) + 3 \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$

(A) I. only (B) II. only
(C) Both I. and II. (D) Neither I. nor II.

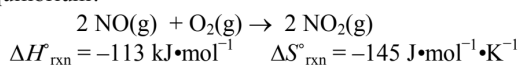
82. What is the value of ΔS° for the reaction below?



Substance	Fe(s)	$\text{Fe}_2\text{O}_3(\text{s})$	CO(g)	$\text{CO}_2(\text{g})$
$S^\circ (\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1})$	27.3	87.4	197.7	213.8

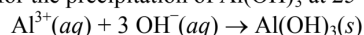
(A) $-44.0 \text{ J}\cdot\text{K}^{-1}$ (B) $-11.8 \text{ J}\cdot\text{K}^{-1}$
(C) $15.5 \text{ J}\cdot\text{K}^{-1}$ (D) $42.8 \text{ J}\cdot\text{K}^{-1}$

83. What is the temperature at which the reaction below is at equilibrium?



(A) -195°C (B) 77.9°C
(C) 506°C (D) 779°C

84. The K_{sp} for $\text{Al}(\text{OH})_3$ is 2.0×10^{-31} . What is the value of ΔG° for the precipitation of $\text{Al}(\text{OH})_3$ at 25°C ?



(A) $-175 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-14.7 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $14.7 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $175 \text{ kJ}\cdot\text{mol}^{-1}$

85. Diethyl ether has a normal boiling point of 34.6°C and a boiling point of -1.5°C at 100 mm Hg. What is the value of $\Delta H^\circ_{\text{vaporization}}$ in $\text{kJ}\cdot\text{mol}^{-1}$?

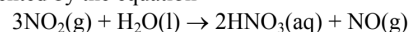
(A) 33.4 (B) 39.1 (C) 64.2 (D) 72.7

86. An ice cube at an unknown temperature is added to 25.0 g of liquid H_2O at 40.0°C . The final temperature of the 29.3 g equilibrated mixture is 21.5°C . What was the original temperature of the ice cube?

$[C_p (\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}) \text{ water} = 4.184, \text{ ice} = 2.06, \Delta H^\circ_{\text{fusion}} = 333 \text{ J/g}]$

(A) -6.5°C (B) -13.1°C
(C) -35.3°C (D) -56.8°C

87. One of the steps in the manufacture of nitric acid is represented by the equation



for which $\Delta H^\circ = -136.5 \text{ kJ}\cdot\text{mol}^{-1}$. Determine $\Delta H^\circ_{\text{formation}}$ for NO_2 in $\text{kJ}\cdot\text{mol}^{-1}$.

Substance	$\text{NO}_2(\text{g})$	$\text{H}_2\text{O}(\text{l})$	$\text{HNO}_3(\text{aq})$	$\text{NO}(\text{g})$
$\Delta H^\circ_{\text{formation}}$	?	-285.8	-207.0	91.3

(A) 33.2 (B) 99.6 (C) 102.2 (D) 157.0

88. For the reaction, $2\text{H}(\text{g}) \rightarrow \text{H}_2(\text{g})$, what are the signs of ΔH° and ΔS° ?

(A) $\Delta H^\circ < 0, \Delta S^\circ < 0$ (B) $\Delta H^\circ < 0, \Delta S^\circ > 0$
(C) $\Delta H^\circ > 0, \Delta S^\circ > 0$ (D) $\Delta H^\circ > 0, \Delta S^\circ < 0$

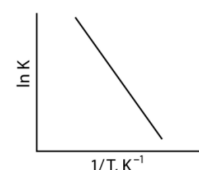
89. Which substance has the greatest molar entropy at 298 K?

(A) $\text{NO}_2(\text{g})$ (B) $\text{N}_2\text{O}_4(\text{l})$
(C) $\text{N}_2\text{O}_4(\text{g})$ (D) $\text{N}_2\text{O}_5(\text{s})$

90. For the process, $\text{CH}_3\text{OH}(\text{l}) \rightarrow \text{CH}_3\text{OH}(\text{g})$ $\Delta G^\circ = 4.30 \text{ kJ}\cdot\text{mol}^{-1}$ at 25°C . What is the vapor pressure of $\text{CH}_3\text{OH}(\text{l})$ at 25°C in mm Hg?

(A) 0.176 mm Hg (B) 14.0 mm Hg
(C) 134 mm Hg (D) 759 mm Hg

91. What quantity is represented by the slope of the line in this graph of the temperature dependence of the natural log of an equilibrium constant?



(A) $-\Delta G^\circ$ (B) $-\Delta G^\circ/R$
(C) $-\Delta H^\circ$ (D) $-\Delta H^\circ/R$

Thermodynamics

92. Spontaneous reactions always
 (A) go to completion.
 (B) are fast.
 (C) involve phase changes.
 (D) release energy and/or show an increase in the system's entropy.
93. The standard enthalpies of combustion of formaldehyde, $\text{H}_2\text{C}=\text{O}(\text{g})$, and formic acid, $\text{HCOOH}(\text{l})$, are -563 and $-270 \text{ kJ}\cdot\text{mol}^{-1}$ respectively. What is ΔH° for the following reaction?
 $\text{H}_2\text{C}=\text{O}(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{HCOOH}(\text{l})$
 (A) $-833 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-293 \text{ kJ}\cdot\text{mol}^{-1}$
 (C) $293 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $833 \text{ kJ}\cdot\text{mol}^{-1}$
94. What is ΔH° for the reaction below?
 $\text{TiCl}_4(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) \rightarrow \text{TiO}_2(\text{s}) + 4 \text{HCl}(\text{g})$
- | Species | $\text{TiCl}_4(\text{g})$ | $\text{H}_2\text{O}(\text{l})$ | $\text{TiO}_2(\text{s})$ | $\text{HCl}(\text{g})$ |
|---|---------------------------|--------------------------------|--------------------------|------------------------|
| $\Delta H^\circ_f \text{ kJ}\cdot\text{mol}^{-1}$ | -763 | -286 | -945 | -92 |
- (A) $298 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $22 \text{ kJ}\cdot\text{mol}^{-1}$
 (C) $12 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $-264 \text{ kJ}\cdot\text{mol}^{-1}$
95. For Br_2 , $\Delta H^\circ_{\text{vap}} = 31 \text{ kJ}\cdot\text{mol}^{-1}$. If S° values for $\text{Br}_2(\text{g})$ and $\text{Br}_2(\text{l})$ are $245 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ and $153 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ respectively, what is the normal boiling point for $\text{Br}_2(\text{l})$?
 (A) 340 K (B) 200 K (C) 130 K (D) 70 K
96. When MgO reacts with H_2O at 25°C and 1 atm, the volume change is $-4.6 \text{ mL}\cdot\text{mol}^{-1}$.
 $\text{MgO}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Mg}(\text{OH})_2(\text{s})$
 What is the value of $\Delta H - \Delta E$ for this reaction?
 (A) $-4.7 \times 10^{-1} \text{ J}\cdot\text{mol}^{-1}$ (B) $-4.7 \times 10^2 \text{ J}\cdot\text{mol}^{-1}$
 (C) $4.7 \times 10^2 \text{ J}\cdot\text{mol}^{-1}$ (D) $4.7 \times 10^{-1} \text{ J}\cdot\text{mol}^{-1}$
97. For the dissolution of Ag_2SO_4 , $\Delta H^\circ = 17.8 \text{ kJ}\cdot\text{mol}^{-1}$ and $\Delta S^\circ = -34.9 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ at 25°C . What is the value of the K_{sp} for Ag_2SO_4 at this temperature?
 (A) 5.0×10^{-2} (B) 7.6×10^{-4}
 (C) 5.3×10^{-4} (D) 1.1×10^{-5}
98. For the reaction:
 $3 \text{H}_2 + \text{N}_2 \rightarrow 2 \text{NH}_3$ $\Delta H^\circ = -97 \text{ kJ}\cdot\text{mol}^{-1}$.
 The H_2 and N_2 bond energies are 436 and 941 $\text{kJ}\cdot\text{mol}^{-1}$, respectively. What is the bond energy of a single N-H bond in $\text{kJ}\cdot\text{mol}^{-1}$?
 (A) 246 (B) 359 (C) 391 (D) 782
99. Calculate ΔE when one mole of liquid is vaporized at its boiling point (80°C) and 1 atm pressure.
 $[\Delta H_{\text{vap}} = 30.7 \text{ kJ/mol}]$
 (A) 33.6 kJ (B) 31.4 kJ (C) 30.0 kJ (D) 27.8 kJ
100. Use the following data to calculate the molar enthalpy of combustion of ethane, C_2H_6 .
 $2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ $\Delta H = -2511 \text{ kJ/mol}$
 $\text{C}_2\text{H}_2(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$ $\Delta H = -311 \text{ kJ/mol}$
 $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$ $\Delta H = -484 \text{ kJ/mol}$
 (A) -1428 kJ/mol (B) -2684 kJ/mol
 (C) -2856 kJ/mol (D) -3306 kJ/mol
101. A 10.00 g piece of metal is heated to 80.00°C and placed in 100.0 g of water at 23.00°C . When the system has reached equilibrium the temperature of the water and metal are 23.50°C . What is the identity of the metal? [Specific heat capacity of $\text{H}_2\text{O} = 4.184 \text{ J/g}\cdot^\circ\text{C}$]
 (A) Ag ($C_p 0.236 \text{ J/g}\cdot^\circ\text{C}$) (B) Cu ($C_p 0.385 \text{ J/g}\cdot^\circ\text{C}$)
 (C) Fe ($C_p 0.449 \text{ J/g}\cdot^\circ\text{C}$) (D) Al ($C_p 0.901 \text{ J/g}\cdot^\circ\text{C}$)
102. For a reaction at constant pressure to be spontaneous, which relationship must be correct?
 (A) $\Delta H_{\text{rxn}} < 0$ (B) $\Delta G_{\text{rxn}} < 0$
 (C) $\Delta S_{\text{rxn}} < 0$ (D) $\Delta S_{\text{univ}} < 0$
103. Tungsten is obtained commercially by the reduction of WO_3 with H_2 according to the equation:
 $\text{WO}_3(\text{s}) + 3 \text{H}_2(\text{g}) \rightarrow \text{W}(\text{s}) + 3 \text{H}_2\text{O}(\text{g})$
 The following data related to this reaction at 25°C are available.
- | | $\text{WO}_3(\text{s})$ | $\text{H}_2\text{O}(\text{g})$ |
|---------------------------------|-------------------------|--------------------------------|
| $\Delta H^\circ \text{ kJ/mol}$ | -840.3 | -241.8 |
| $\Delta G^\circ \text{ kJ/mol}$ | -763.5 | -228.5 |
- The temperature at which this reaction is at equilibrium at 1 atm is closest to which of the following?
 (A) 124 K (B) 213 K (C) 928 K (D) 2810 K
104. The gaseous compound NOBr decomposes according to the equation
 $\text{NOBr}(\text{g}) \rightleftharpoons \text{NO}(\text{g}) + 1/2 \text{Br}_2(\text{g})$
 At 350 K the equilibrium constant, K_p , is 0.15. What is the value of ΔG° ?
 (A) $-5.5 \times 10^3 \text{ J/mol}$ (B) $-2.4 \times 10^3 \text{ J/mol}$
 (C) $2.4 \times 10^3 \text{ J/mol}$ (D) $5.5 \times 10^3 \text{ J/mol}$
105. Which statement is correct at 25°C and 1 atm pressure?
 (A) ΔG°_f for $\text{Hg}(\text{l}) = 0 \text{ kJ}\cdot\text{mol}^{-1}$
 (B) ΔH°_f for $\text{I}_2(\text{g}) = 0 \text{ kJ}\cdot\text{mol}^{-1}$
 (C) ΔH°_f for $\text{H}_2\text{O}(\text{l}) = 0 \text{ kJ}\cdot\text{mol}^{-1}$
 (D) S° for $\text{O}_2(\text{g}) = 0 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$

Thermodynamics

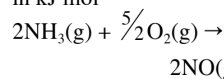
106. What is the relationship among the magnitudes of the enthalpies of combustion (ΔH_{comb}), fusion (ΔH_{fus}) and vaporization (ΔH_{vap}) for a hydrocarbon such as hexane, C_6H_{14} ?

(A) $\Delta H_{\text{fus}} < \Delta H_{\text{comb}} < \Delta H_{\text{vap}}$
 (B) $\Delta H_{\text{vap}} < \Delta H_{\text{fus}} < \Delta H_{\text{comb}}$
 (C) $\Delta H_{\text{comb}} < \Delta H_{\text{vap}} < \Delta H_{\text{fus}}$
 (D) $\Delta H_{\text{fus}} < \Delta H_{\text{vap}} < \Delta H_{\text{comb}}$

107. Which choice represents the signs for ΔS and ΔH for the sublimation of a compound that is occurring at constant temperature?

(A) $\Delta S < 0$, $\Delta H < 0$ (B) $\Delta S < 0$, $\Delta H > 0$
 (C) $\Delta S > 0$, $\Delta H > 0$ (D) $\Delta S > 0$, $\Delta H < 0$

108. Determine ΔH_{rxn} for this reaction in $\text{kJ}\cdot\text{mol}^{-1}$



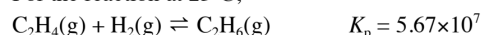
$\Delta H_f^\circ, \text{kJ}\cdot\text{mol}^{-1}$	
$\text{H}_2\text{O}(\text{g})$	-241.8
$\text{NH}_3(\text{g})$	-46.1
$\text{NO}(\text{g})$	90.3

(A) -105.4 (B) -226.3 (C) -452.6 (D) -637.0

109. The enthalpy of fusion for $\text{NaF}(\text{s})$ at its melting point (992°C) is $29.3 \text{ kJ}\cdot\text{mol}^{-1}$. What is the value of $\Delta S_{\text{fus}}^\circ$ in $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$?

(A) 43.2 (B) 33.9 (C) 29.5 (D) 23.2

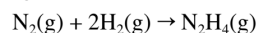
110. For the reaction at 25°C ,



What is ΔG° for this reaction in $\text{kJ}\cdot\text{mol}^{-1}$?

(A) -0.436 (B) -3.71 (C) -19.2 (D) -44.2

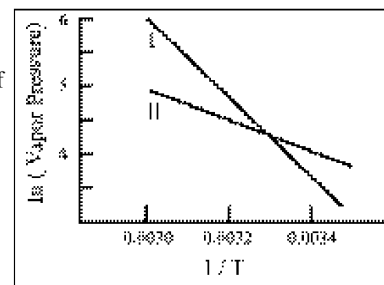
111. Given the bond energies, what is the ΔH_f° $\text{N}_2\text{H}_4(\text{g})$ in $\text{kJ}\cdot\text{mol}^{-1}$?



Bond Energies, $\text{kJ}\cdot\text{mol}^{-1}$	
$\text{N}\equiv\text{N}$	946
$\text{N}=\text{N}$	418
$\text{N}-\text{N}$	163
$\text{N}-\text{H}$	389
$\text{H}-\text{H}$	436

(A) 156 (B) 99 (C) -99 (D) -156

112. In the graph, the natural log of the vapor pressures of two substances are plotted versus $1/T$. What can be concluded about the relative enthalpies of vaporization (ΔH_{vap}) of these substances?



(A) ΔH_{vap} of I is greater than ΔH_{vap} of II
 (B) ΔH_{vap} of I is less than ΔH_{vap} of II
 (C) ΔH_{vap} of I is equal to ΔH_{vap} of II
 (D) No conclusion can be drawn from this information alone.

113. Use the information provided to calculate the standard enthalpy of formation of acetylene, $\text{C}_2\text{H}_2(\text{g})$, in $\text{kJ}\cdot\text{mol}^{-1}$.

$\text{C}_2\text{H}_2(\text{g}) + 5/2\text{O}_2(\text{g})$	$\rightarrow 2\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	$\Delta H^\circ = -1299.5 \text{ kJ}$
$\text{C}(\text{s}) + \text{O}_2(\text{g})$	$\rightarrow \text{CO}_2(\text{g})$	$\Delta H^\circ = -393.5 \text{ kJ}$
$\text{H}_2(\text{g}) + 1/2\text{O}_2(\text{g})$	$\rightarrow \text{H}_2\text{O}(\text{l})$	$\Delta H^\circ = -285.8 \text{ kJ}$

(A) -1978.8 (B) -1121.4
 (C) 226.7 (D) 453.4

114. Which statement is always true for a spontaneous reaction?

(A) The entropy change for the system is negative.
 (B) The enthalpy change for the system is negative.
 (C) The entropy change for the universe is positive.
 (D) The free energy change for the system is positive.

115. The heat of a reaction is measured in a bomb calorimeter. This heat is equal to which thermodynamic quantity?

(A) ΔE (B) ΔG (C) ΔH (D) ΔS

116. 84.12 g of gold at 120.1°C is placed in 106.4 g of H_2O at 21.4°C . What is the final temperature of this system?

Specific heat capacities / $\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$	
$\text{Au}(\text{s})$	0.129
$\text{H}_2\text{O}(\text{l})$	4.184

(A) 70.8 (B) 65.0 (C) 27.8 (D) 23.7

117. In order to calculate the lattice energy of NaCl using a Born-Haber cycle, which value is not needed?

(A) enthalpy of sublimation of $\text{Na}(\text{s})$
 (B) first ionization energy of $\text{Cl}(\text{g})$
 (C) bond dissociation energy of $\text{Cl}_2(\text{g})$
 (D) enthalpy of formation of $\text{NaCl}(\text{s})$

Thermodynamics

118. Liquid bromine boils at 332.7 K. Estimate the enthalpy of formation of $\text{Br}_2(\text{g})$ in $\text{kJ}\cdot\text{mol}^{-1}$.
- | $S^\circ / \text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ | |
|--|------|
| $\text{Br}_2(\text{g})$ | 58.6 |
| $\text{Br}_2(\text{l})$ | 36.4 |
- (A) 7.40 (B) 12.1 (C) 19.5 (D) 22.2

119. The vapor pressure of phosphorus trichloride is 100 mm Hg at 21.0°C and its normal boiling point is 74.2°C. What is its enthalpy of vaporization in $\text{kJ}\cdot\text{mol}^{-1}$?
- (A) 0.493 (B) 3.93 (C) 23.0 (D) 32.4

120. Calculate the standard enthalpy of formation of acetylene (in $\text{kJ}\cdot\text{mol}^{-1}$).
- $$2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \quad \Delta H^\circ = -2243.6 \text{ kJ}$$
- $$\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \quad \Delta H^\circ = -393.5 \text{ kJ}$$
- $$\text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}) \quad \Delta H^\circ = -285.8 \text{ kJ}$$
- (A) 49.0 (B) 98.0 (C) 1121.8 (D) 1564.3

121. The boiling point of diethyl ether is 34.6°C. Which is true for the vaporization of diethyl ether at 25.0°C?
- (A) $\Delta G^\circ_{\text{vap}} > 0$ (B) $\Delta H^\circ_{\text{vap}} < 0$
 (C) $K_{\text{vap}} = 1$ (D) $\Delta S^\circ_{\text{vap}} < 0$

122. Estimate the enthalpy of combustion of methane in $\text{kJ}\cdot\text{mol}^{-1}$.
- | Bond Dissociation Enthalpies / $\text{kJ}\cdot\text{mol}^{-1}$ | | | |
|--|-----|-----|-----|
| C–C | 350 | C–O | 350 |
| C–H | 410 | C=O | 732 |
| O–H | 460 | O–O | 180 |
| | | O=O | 498 |
- $$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$$
- (A) 668 (B) 540 (C) -540 (D) -668

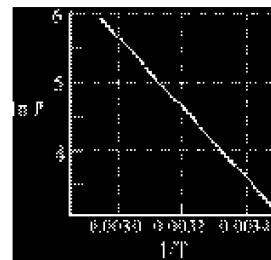
123. Which reaction has a positive $\Delta S^\circ_{\text{reaction}}$?

- (A) $\text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$
 (B) $2\text{C}_2\text{H}_6(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$
 (C) $\text{N}_2(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow \text{N}_2\text{H}_4(\text{g})$
 (D) $2\text{H}_2\text{O}_2(\text{l}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

124. For reactions conducted at constant pressure, under what conditions are ΔE and ΔH equal?
- | |
|-----------------------------|
| I. constant number of moles |
| II. constant temperature |
| III. constant volume |
- (A) I only (B) II only
 (C) III only (D) I and II only

125. For the reaction,
- $$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$$
- $K_p = 50.0$ at 721 K. What is the value of ΔG° for this reaction (per mole of H_2) at 721 K?
- (A) -32.3 kJ (B) -23.5 kJ
 (C) -10.2 kJ (D) -0.231 kJ

126. What is ΔH_{vap} for the substance whose vapor pressure is represented by the diagram?



- (A) 4.8 $\text{kJ}\cdot\text{mol}^{-1}$ (B) 33 $\text{kJ}\cdot\text{mol}^{-1}$
 (C) 44 $\text{kJ}\cdot\text{mol}^{-1}$ (D) 50 $\text{kJ}\cdot\text{mol}^{-1}$

127. For a reaction that is exothermic and non-spontaneous at 25 °C, which quantity must be positive?

- (A) ΔE° (B) ΔG° (C) ΔH° (D) ΔS°

128. Use the thermochemical data given to calculate ΔH° for $\text{N}_2\text{O}_5(\text{g})$ in $\text{kJ}\cdot\text{mol}^{-1}$.

- $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) \quad \Delta H^\circ = +180.5 \text{ kJ}$
 $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}) \quad \Delta H^\circ = -114.1 \text{ kJ}$
 $4\text{NO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{N}_2\text{O}_5(\text{g}) \quad \Delta H^\circ = -110.2 \text{ kJ}$

- (A) -332.8 (B) -43.8 (C) 11.3 (D) 22.6

129. Bromine boils at 59°C with $\Delta H^\circ_{\text{vap}} = 29.6 \text{ kJ}\cdot\text{mol}^{-1}$. What is the value of $\Delta S^\circ_{\text{vap}}$ in $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$?

- (A) 11.2 (B) 89.2 (C) 501 (D) 1750

130. The K_{sp} of calcium fluoride is 3.2×10^{-11} . Calculate the ΔG° (in $\text{kJ}\cdot\text{mol}^{-1}$) for the dissolving of solid calcium fluoride at 25°C.

- (A) 2.18 (B) 5.02 (C) 26.0 (D) 59.9

131. For which exothermic reaction is ΔE more negative than ΔH ?

- (A) $\text{Br}_2(\text{l}) \rightleftharpoons \text{Br}_2(\text{g})$
 (B) $2\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$
 (C) $\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \rightleftharpoons 2\text{HF}(\text{g})$
 (D) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$

132. For a reaction at 25°C, $\Delta G = 12.7 \text{ kJ}$ when the reaction quotient, $Q = 10.0$. What is the value of ΔG° for this reaction?

- (A) -12.1 kJ (B) 7.0 kJ
 (C) 18.4 kJ (D) 37.5 kJ

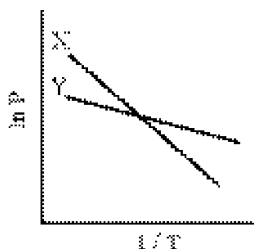
133. Which terms are exothermic for the formation of $\text{NaF}(\text{s})$?

- | |
|---|
| I. $\text{Na}(\text{g}) \rightleftharpoons \text{Na}^+(\text{g}) + \text{e}^-$ |
| II. $\text{F}(\text{g}) + \text{e}^- \rightleftharpoons \text{F}^-(\text{g})$ |
| III. $\text{Na}^+(\text{g}) + \text{F}^-(\text{g}) \rightleftharpoons \text{NaF}(\text{s})$ |

- (A) I only (B) II only
 (C) I and III only (D) II and III only

Thermodynamics

134. According to the graph (ln vapor pressure vs $1/T$) what can be concluded about the enthalpies of vaporization (ΔH_{vap}) of liquids X and Y?



- (A) $\Delta H_{\text{vap}} X > \Delta H_{\text{vap}} Y$
 (B) $\Delta H_{\text{vap}} X = \Delta H_{\text{vap}} Y$
 (C) $\Delta H_{\text{vap}} X < \Delta H_{\text{vap}} Y$
 (D) No conclusions can be drawn about the relative ΔH_{vap} values from this diagram.

135. Calculate ΔH° for the reaction;
 $\text{TiCl}_4(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{TiO}_2(\text{s}) + 4\text{HCl}(\text{g})$

ΔH_f° kJ/mol	
-763	$\text{TiCl}_4(\text{g})$
-286	$\text{H}_2\text{O}(\text{l})$
-945	$\text{TiO}_2(\text{s})$
-92	$\text{HCl}(\text{g})$

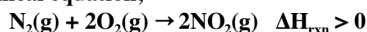
- (A) -264 kJ (B) 12 kJ
 (C) 22 kJ (D) 298 kJ

136. Use bond energies to estimate the value of ΔH° for the reaction;
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$

Bond Energies kJ/mol	
436	H-H
386	H-N
193	N-N
418	N=N
941	N≡N

- (A) -995 kJ (B) -590 kJ
 (C) -67 kJ (D) 815 kJ

Questions 137, and 138, should be answered using this thermochemical equation;



137. Which relationship is correct for this reaction at a pressure of 1 atm?
- (A) $\Delta E_{\text{rxn}} > \Delta H_{\text{rxn}}$ (B) $\Delta E_{\text{rxn}} < \Delta H_{\text{rxn}}$
 (C) $\Delta E_{\text{rxn}} = \Delta H_{\text{rxn}} + \Delta S_{\text{rxn}}$ (D) $\Delta E_{\text{rxn}} = \Delta H_{\text{rxn}} - \Delta S_{\text{rxn}}$
138. Under what temperature conditions is this reaction spontaneous at standard pressure?
- (A) at low temperatures only
 (B) at high temperatures only
 (C) at all temperatures
 (D) at no temperature

139. Diethyl ether has a normal boiling point of 35.0°C and has an entropy of vaporization of $84.4 \text{ J/mol}\cdot\text{K}$. What is its enthalpy of vaporization?

- (A) 0.274 J/mol (B) 2.41 J/mol
 (C) 3.65 J/mol (D) 26.0 kJ/mol

140. A 9.40 g sample of KBr is dissolved in 105 g of H_2O at 23.6°C in a coffee cup. Find the

Solution Properties	
Molar mass KBr	119 g/mol
ΔH_{soln} KBr	19.9 kJ/mol
C_p solution	$4.184 \text{ J/g}\cdot^\circ\text{C}$

final temperature of this system. Assume that no heat is transferred to the cup or the surroundings.

- (A) 20.0°C (B) 20.3°C
 (C) 26.9°C (D) 27.2°C

141. For a substance with the values of ΔH_{vap} and ΔS_{vap} given below, what is its normal boiling point in $^\circ\text{C}$? ($\Delta H_{\text{vap}} = 59.0 \text{ kJ}\cdot\text{mol}^{-1}$; $\Delta S_{\text{vap}} = 93.65 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$)

- (A) 357 (B) 630 (C) 1314 (D) 1587

142. For which reaction is ΔH (enthalpy change) most nearly equal to ΔE (internal energy change)?

- (A) $\text{H}_2(\text{g}) + 1/2\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$
 (B) $\text{Cl}_2(\text{g}) + \text{F}_2(\text{g}) \rightarrow 2\text{ClF}(\text{g})$
 (C) $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$
 (D) $2\text{SO}_3(\text{g}) \rightarrow 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$

143. Which is the best description of the relationship between the absolute entropies, S° , of solid water at 100 K and at 200 K?

- (A) $S^\circ_{200\text{K}}$ is smaller because entropy decreases as temperature increases.
 (B) $S^\circ_{200\text{K}}$ is smaller because the surroundings are more disordered at higher temperatures.
 (C) $S^\circ_{100\text{K}} = S^\circ_{200\text{K}}$ because water is in the solid phase at both temperatures.
 (D) $S^\circ_{200\text{K}}$ is larger because the vibration of the molecules increases as temperature increases.

144. For the reaction,
 $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$
 which expression gives ΔH° ?

Bond dissociation energies	$\text{kJ}\cdot\text{mol}^{-1}$
C-H	413
C-Cl	328
Cl-Cl	242
H-Cl	431

- (A) $\Delta H = (413 + 328) - (242 + 431)$
 (B) $\Delta H = (413 - 328) - (242 - 431)$
 (C) $\Delta H = (413 - 242) - (328 - 431)$
 (D) $\Delta H = (413 + 242) - (328 + 431)$

Thermodynamics

145. Which phase change for water has positive values for both ΔH° and ΔG° ?

(A) (l) $\rightarrow$ (s) at 250 K (B) (l) $\rightarrow$ (s) at 350 K
(C) (l) $\rightarrow$ (g) at 350 K (D) (l) $\rightarrow$ (g) at 450 K

146. When solid CuSO_4 dissolves in water to make a 1M solution, the temperature of the system increases. When solid NH_4NO_3 dissolves in water to make a 1 M solution, the temperature of the system decreases. Which statement(s) must be correct for these dissolving processes?

I ΔH° values for both processes have the same sign.
II ΔG° values for both processes have the same sign.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

147. A liquid has a vapor pressure of 40 mmHg at 19.0 °C and a normal boiling point of 78.3 °C. What is its enthalpy of vaporization in $\text{kJ}\cdot\text{mol}^{-1}$?

(A) 42.4 (B) 18.4 (C) 5.10 (D) 1.45

148. When the substances below are arranged in order of increasing entropy values, S° , at 25 °C which is the correct order?

(A) $\text{CO}_2(\text{s}) < \text{CO}_2(\text{aq}) < \text{CO}_2(\text{g})$
(B) $\text{CO}_2(\text{g}) < \text{CO}_2(\text{aq}) < \text{CO}_2(\text{s})$
(C) $\text{CO}_2(\text{s}) < \text{CO}_2(\text{g}) < \text{CO}_2(\text{aq})$
(D) $\text{CO}_2(\text{g}) < \text{CO}_2(\text{s}) < \text{CO}_2(\text{aq})$

149. When 50. mL of 0.10 M HCl is mixed with 50. mL of 0.10 M NaOH the temperature of the solution increases by 3.0 °C. Calculate the $\Delta H_{\text{neutralization}}$ per mole of HCl. (The solution has a density = $1.0 \text{ g}\cdot\text{mL}^{-1}$ and $C_p = 4.2 \text{ J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$)

(A) $1.3 \times 10^3 \text{ kJ}$ (B) $-1.3 \times 10^2 \text{ kJ}$
(C) $-2.5 \times 10^2 \text{ kJ}$ (D) $-1.3 \times 10^3 \text{ kJ}$

150. The combustion of 0.200 mol of liquid carbon disulfide, CS_2 , to give $\text{CO}_2(\text{g})$ and $\text{SO}_2(\text{g})$ releases 215 kJ of heat. What is ΔH_f° for $\text{CS}_2(\text{l})$ in $\text{kJ}\cdot\text{mol}^{-1}$?

ΔH_f°	$\text{kJ}\cdot\text{mol}^{-1}$
$\text{CO}_2(\text{g})$	-393.5
$\text{SO}_2(\text{g})$	-296.8

(A) 385 (B) 87.9 (C) -385 (D) -475

151. For the reaction: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ $\Delta H < 0$. What predictions can be made about the sign of ΔS and the temperature conditions under which the reaction would be spontaneous?

ΔS_{rxn}	Temperature Condition
(A) negative	low temperatures
(B) negative	high temperatures
(C) positive	high temperatures

152. As ΔG° for a reaction changes from a large negative value to a large positive value, K for the reaction will change from

(A) a large positive value to a large negative value.
(B) a large positive value to a small positive value.
(C) a large negative value to a large positive value.
(D) a large negative value to a small negative value.

153. ΔE° is measured at constant volume and ΔH° is measured at constant pressure. For the reaction;



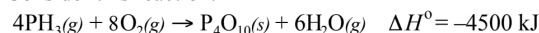
How do the ΔE° and ΔH° compare for this reaction?

(A) $\Delta E^\circ < \Delta H^\circ$ (B) $\Delta E^\circ > \Delta H^\circ$
(C) $\Delta E^\circ = \Delta H^\circ$ (D) Impossible to tell from this information.

154. Which statement is correct?

(A) In a coffee-cup calorimeter, $q = \Delta H$.
(B) In a coffee-cup calorimeter, $w = 0$.
(C) In a bomb calorimeter, $q = \Delta S$.
(D) In a bomb calorimeter, $w > 0$.

155. Consider this reaction.



Calculate ΔH_f° of $\text{P}_4\text{O}_{10}(\text{s})$ in $\text{kJ}\cdot\text{mol}^{-1}$.

Substance	ΔH_f° , $\text{kJ}\cdot\text{mol}^{-1}$
$\text{PH}_3(\text{g})$	+9.2
$\text{H}_2\text{O}(\text{g})$	-241.8

(A) -5914 kJ (B) -4751 kJ
(C) -4249 kJ (D) -3012 kJ

156. For which substances and conditions can $S^\circ = 0$?

I. elements at 0 K
II. compounds at 0 K
III. gases at 298 K

(A) I only (B) III only
(C) I and II only (D) I and III only

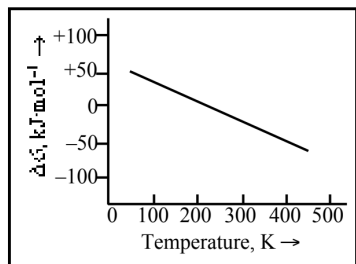
Thermodynamics

157. 50.0 mL of 0.10 M HCl is mixed with 50.0 mL of 0.10 M NaOH. The solution temperature rises by 3.0 °C. Calculate the enthalpy of neutralization per mole of HCl.

Solution Values	
C_p	$4.18 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1}$
density	$1.0 \text{ g} \cdot \text{mL}^{-1}$

- (A) $-2.5 \times 10^2 \text{ kJ}$ (B) $-1.3 \times 10^2 \text{ kJ}$
(C) $-8.4 \times 10^1 \text{ kJ}$ (D) $-6.3 \times 10^1 \text{ kJ}$

158. What can be concluded about the values of ΔH and ΔS from this graph?



- (A) $\Delta H > 0$, $\Delta S > 0$ (B) $\Delta H > 0$, $\Delta S < 0$
(C) $\Delta H < 0$, $\Delta S > 0$ (D) $\Delta H < 0$, $\Delta S < 0$

159. The boiling point of chloroform, CHCl_3 , is 61.7 °C and its enthalpy of vaporization is $31.4 \text{ kJ} \cdot \text{mol}^{-1}$. Calculate the molar entropy of vaporization for chloroform.

- (A) $10.7 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ (B) $93.8 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$
(C) $301 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ (D) $509 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

160. ΔG° for a reaction at 25 °C is $30.5 \text{ kJ} \cdot \text{mol}^{-1}$. What is the value of K ?

- (A) 2.2×10^5 (B) 1.1
(C) 0.86 (D) 4.5×10^{-6}

161. How much energy must be supplied to change 36 g of ice at 0 °C to water at room temperature, 25 °C?

Data for Water, H_2O	
$\Delta H^\circ_{\text{fusion}}$	$6.01 \text{ kJ} \cdot \text{mol}^{-1}$
$C_{p, \text{liquid}}$	$4.18 \text{ J} \cdot \text{K}^{-1} \cdot \text{g}^{-1}$

- (A) 12 kJ (B) 16 kJ
(C) 19 kJ (D) 22 kJ

162. For a process that is both endothermic and spontaneous,

- (A) $\Delta H < 0$ (B) $\Delta G > 0$
(C) $\Delta E = 0$ (D) $\Delta S > 0$

163. Consider the values for ΔH° (in $\text{kJ} \cdot \text{mol}^{-1}$) and for ΔS° (in $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$) given for four different reactions. For which reaction will ΔG° increase the most (becoming more positive) when the temperature is increased from 0 °C to 25 °C?

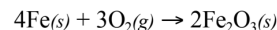
- (A) $\Delta H^\circ = 50$, $\Delta S^\circ = 50$
(B) $\Delta H^\circ = 90$, $\Delta S^\circ = 20$
(C) $\Delta H^\circ = -20$, $\Delta S^\circ = -50$
(D) $\Delta H^\circ = -90$, $\Delta S^\circ = -20$

164. A certain reaction is exothermic by 220 kJ and does 10 kJ of work. What is the change in the internal energy of the system at constant temperature?

- (A) +230 kJ (B) +210 kJ
(C) -210 kJ (D) -230 kJ

165. $\text{Fe}_2\text{O}_3(s) + 3/2\text{C}(s) \rightarrow 3/2\text{CO}_2(g) + 2\text{Fe}(s)$ $\Delta H^\circ = +234.1 \text{ kJ}$
 $\text{C}(s) + \text{O}_2(g) \rightarrow \text{CO}_2(g)$ $\Delta H^\circ = -393.5 \text{ kJ}$

Use these equations and ΔH° values to calculate ΔH° for this reaction.



- (A) -1648.7 kJ (B) -1255.3 kJ
(C) -1021.2 kJ (D) -129.4 kJ

166. A large positive value of ΔG° corresponds to which of these?

- (A) small positive K (B) small negative K
(C) large positive K (D) large negative K

167. A bomb calorimeter has a heat capacity of $783 \text{ J} \cdot ^\circ\text{C}^{-1}$ and contains 254 g of water, which has a specific heat of $4.184 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1}$. How much heat is evolved or absorbed by a reaction when the temperature goes from 23.73 °C to 26.01 °C?

- (A) 1.78 kJ absorbed (B) 2.42 kJ absorbed
(C) 1.78 kJ evolved (D) 4.21 kJ evolved

168. Consider this equation and the associated value for ΔH° .

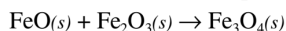


Which statement about this information is *incorrect*?

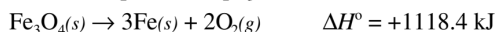
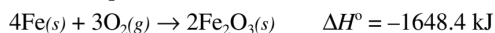
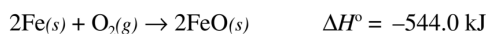
- (A) If the equation is reversed, the ΔH° value equals +92.3 kJ.
(B) The four HCl bonds are stronger than the four bonds in H_2 and Cl_2 .
(C) The ΔH° value will be -92.3 kJ if the HCl is produced as a liquid.
(D) 23.1 kJ of heat will be evolved when 1 mol of $\text{HCl}(g)$ is produced.

Thermodynamics

169. Determine the heat of reaction for this process.



Given information:

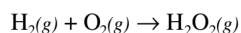


- (A) -1074.0 kJ (B) -22.2 kJ
(C) $+249.8 \text{ kJ}$ (D) $+2214.6 \text{ kJ}$

170. For which process will ΔH° and ΔG° be expected to be most similar?

- (A) $2\text{Al}(s) + \text{Fe}_2\text{O}_3(s) \rightarrow 2\text{Fe}(s) + \text{Al}_2\text{O}_3(s)$
(B) $2\text{Na}(s) + 2\text{H}_2\text{O}(l) \rightarrow 2\text{NaOH}(aq) + \text{H}_2(g)$
(C) $2\text{NO}_2(g) \rightarrow \text{N}_2\text{O}_4(g)$
(D) $2\text{H}_2(g) + \text{O}_2(g) \rightarrow 2\text{H}_2\text{O}(g)$

171. Use bond energies to estimate ΔH for this reaction.



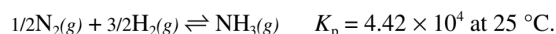
Bond	Bond Energy
H-H	436 kJ·mol ⁻¹
O-O	142 kJ·mol ⁻¹
O=O	499 kJ·mol ⁻¹
H-O	460 kJ·mol ⁻¹

- (A) -127 kJ (B) -209 kJ
(C) -484 kJ (D) -841 kJ

172. For a particular reaction, $\Delta H^\circ = -38.3 \text{ kJ}$ and $\Delta S^\circ = -113 \text{ J·K}^{-1}$. This reaction is

- (A) spontaneous at all temperatures.
(B) nonspontaneous at all temperatures.
(C) spontaneous at temperatures below 66°C .
(D) spontaneous at temperatures above 66°C .

173. What is ΔG° for this reaction?



- (A) $-26.5 \text{ kJ·mol}^{-1}$ (B) $-11.5 \text{ kJ·mol}^{-1}$
(C) -2.2 kJ·mol^{-1} (D) $-0.97 \text{ kJ·mol}^{-1}$

174. $3\text{N}_2\text{O}(g) + 2\text{NH}_3(g) \rightarrow 4\text{N}_2(g) + 3\text{H}_2\text{O}(g) \quad \Delta H = -879.6 \text{ kJ}$

What is ΔH_f° for N_2O in kJ·mol^{-1} ?

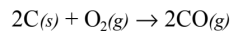
Heats of Formation	
NH_3	$-45.9 \text{ kJ·mol}^{-1}$
H_2O	$-241.8 \text{ kJ·mol}^{-1}$

- (A) $+246$ (B) $+82$ (C) -82 (D) -246

175. What is the change in internal energy, ΔE , for a reaction that gives off 65 joules of heat and does 38 joules of work?

- (A) -103 J (B) -27 J (C) $+27 \text{ J}$ (D) $+103 \text{ J}$

176. What are the signs of ΔH and ΔS for this reaction?



	ΔH	ΔS
(A)	—	—
(B)	—	+
(C)	+	+
(D)	+	—

177. Which statements are true?

- S° values for all elements in their standard states are positive.
- S° values for all aqueous ions are positive.
- ΔS° values for all spontaneous reactions are positive.

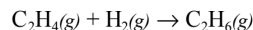
- (A) 1 only (B) 1 and 2 only
(C) 2 and 3 only (D) 1, 2 and 3

178. $\text{Ag}^+(aq) + 2\text{NH}_3(aq) \rightleftharpoons \text{Ag}(\text{NH}_3)_2^+(aq)$

For this reaction, $K = 1.7 \times 10^7$ at 25°C . What is the value of ΔG° in kJ?

- (A) -41.2 (B) -17.9 (C) $+17.9$ (D) $+41.2$

179. The value of ΔH for a reaction can be found by appropriate combination of bond enthalpies (the energy required to break a particular bond, represented BE). Which expression will give ΔH for this reaction?



- (A) $BE_{\text{C=C}} + BE_{\text{H-H}} - [BE_{\text{C-C}} + 2BE_{\text{C-H}}]$
(B) $BE_{\text{C-C}} + 2BE_{\text{C-H}} - [BE_{\text{C=C}} + BE_{\text{H-H}}]$
(C) $1/2BE_{\text{C=C}} + BE_{\text{H-H}} - 2BE_{\text{C-H}}$
(D) $2BE_{\text{C-H}} - 1/2BE_{\text{C=C}} + BE_{\text{H-H}}$